设备名称：探测器 DET8A2

1. 设备图片：



2. 存放地点：8#2014

3. 关键性能参数（文字或者图片快照）：

硅探测器，320 - 1000 nm，上升时间 720 ps，0.5 mm^2

4. 购买链接：

https://www.thorlabschina.cn/item/det8a2

5. 驱动与控制软件、参数手册（包括 PDF 附件）：

简要说明（或者截图）即可，其余手册发送 PDF 附件给我。

1.4 Technical Data

1.4.1 Specifications

Electrical Specifications		
Detector	-	Silicon (Si)
Active Area	-	0.5 mm² (Ø0.8 mm)
Wavelength Range	λ	320 nm to 1000 nm
Peak Wavelength	λ_p	800 nm (Typ.)
Peak Response[a]	$\Re(\lambda_p)$	0.57 A/W (Typ.)
Junction Capacitance[a]	C_J	3 pF (Typ.)
Rise/Fall Time (650 nm)[a,b,c]	t_r	720 ps (Typ.)
NEP[a]	-	3.1 x 10^{-15} W/(Hz$^{1/2}$) (Typ.)
Bias Voltage	V_R	10 V
Dark Current[a,d]	I_D	0.01 nA (Typ.) 0.5 nA (Max)
Output Current	I_{OUT}	0 to 10 mA

a. Measured with a specified bias voltage of 10.0 V.
b. For a 50 Ω load. Low battery voltage will result in slower rise times and decreased bandwidth.
c. Rise/fall time is wavelength dependent. As the wavelength decreases or increases from 650 nm, the rise/fall times will decrease or increase, respectively.
d. Measured with a 1 MΩ Load

General	
On/Off Switch	Slide
Battery Check Switch	Momentary Pushbutton
Output	BNC (DC Coupled)
Package Size	2.79" X 21.96" x 0.89" (70.9 mm x 49.8 mm x 22.5 mm)
PD Surface Depth[a]	0.09" (2.2 mm)
Weight, Detector Only	0.10 kg
Accessories	SM1T1 Coupler SM1RR Retainer Ring
Operating Temp	10 °C to 50 °C
Storage Temp	-20 °C to 70 °C
Battery	A23, 12 V_{DC}, 40 mAh
Low Battery Voltage[b] V_{OUT} **(Hi-Z)** V_{OUT} **(50 Ω)**	(See Section 3.9) ~9 V ~170 mV

a. Measured from the active area to the start of the threads on the housing body. The detector active area surface is flush with the front of the housing body.
b. Assumes the battery voltage drops to 9.6 V. The reverse protection diode generates a 0.6 V drop.

1.4.2 Response Curve

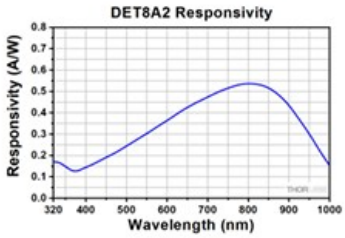


Figure 1 Typical Response Curve

1.4.3 Mechanical Drawings

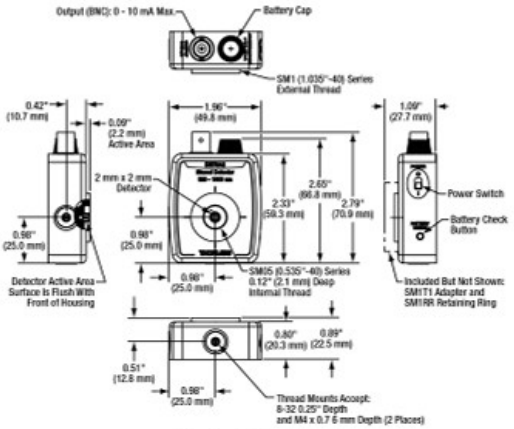


Figure 2 Mechanical Drawing